SIMONE ROMITI

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I am a theoretical physicist specialized in lattice gauge theories, with 4+ years of postdoctoral experience. My research focuses on phenomenological predictions for lattice QCD, and algorithms development for the simulation of strongly-interacting quantum field theories on the lattice. My computational expertise includes high-performance computing for large-scale numerical simulations, Bayesian methods, and machine learning techniques for many-body systems (Physics-Informed Neural Networks, Diffusion Models). I have a proven track record of designing algorithms that delivered order-of-magnitude performance gains, of building open-source scientific software used in peer-reviewed research, and leading collaborative projects across international research institutions.

TECHNICAL SKILLS

Programming Languages - C, C++, Python, Bash, R, SQL
HPC & Systems - CUDA, MPI, OpenMP, SLURM, EasyBuild, GNU/Linux
ML & Data - PINNs, VAEs, Diffusion Models, Bayesian Statistics, Markov Chain Monte Carlo, Nested Sampling
Libraries & Frameworks - PyTorch, NumPy, SciPy, Pandas, Matplotlib, Plotly, Streamlit, SymPy, Jupyter
DevOps & Tools - Git, GitHub Actions, CI/CD, Docker, LaTeX, Quarto, Markdown
Spoken Languages - Italian (native), English (fluent), German (basic)

WORK EXPERIENCE

University of Bern — Apr 2024 – Present
Postdoctoral Research Scientist — Bern, Switzerland

- **Designed novel algorithm for the solution of lattice Hamiltonians** using Physics-Informed Neural Networks (PINNs)
- **Achieved sub-permille numerical precision** as lead developer for a high-accuracy scientific computation of HVP contribution to the muon's $g_\mu - 2$
- **Improved algorithm scaling from $O(N^6)$ to $O(N \log N)$** through a novel mathematical reformulation using FFTs and convolutions, entering in the HLbL contribution to $g_\mu - 2$
- **Built and maintained open-source simulation libraries** (Python/C++) adopted across multiple research groups and directly enabling peer-reviewed publications
- **Mentored 3 PhD students** on software engineering practices, numerical methods, and project delivery

University of Bonn — Nov 2021 – Mar 2024
Postdoctoral Research Scientist — Bonn, Germany

- **Optimized GPU kernels (CUDA)**, achieving a $\sim$1.5$\times$ throughput improvement via auto-tuning of Multigrid solver parameters on HPC clusters
- **Designed and ran large-scale distributed simulations** (MPI/OpenMP) on international computing centers
- **Developed a novel numerical formulation of canonical momenta in $SU(N)$**, achieving machine-precision exactness for a class of operator algebra constraints
- **Integrated Monte Carlo simulations with quantum computing workflows**, delivering end-to-end prototype pipelines bridging classical HPC and quantum backends
- **Mentored 3 graduate students** and delivered tutorial sessions for undergraduate computing courses

SELECTED PUBLICATIONS

- SU(N) lattice gauge theories with Physics-Informed Neural Networks
- The anomalous magnetic moment of the muon in the Standard Model: an update
- Strange and charm quark contributions to the muon anomalous magnetic moment in lattice QCD with twisted-mass fermions
- Towards determining the (2+1)-dimensional Quantum Electrodynamics running coupling with Monte Carlo and quantum computing methods
- Digitizing lattice gauge theories in the magnetic basis: reducing the breaking of the fundamental commutation relations

EDUCATION

Roma Tre University — Rome, Italy
PhD in Physics (Dissertation April 2022) — Nov. 2018 – Oct. 2021

- Thesis: *Non-perturbative aspects of gauge theories on the lattice: algorithms, simulations and quantum computing*
- Focus: Monte Carlo methods, HPC simulations, Isospin Breaking effects in lattice QCD
- Ranked **1st** in national competitive admission exam

Roma Tre University — Rome, Italy
M.S. in Physics — Oct. 2016 – Oct. 2018

- GPA: 29.85/30, graduated with highest honours

Roma Tre University — Rome, Italy
B.S. in Physics — Oct. 2013 – Jul. 2016

- GPA: 28.84/30, graduated with highest honours | Merit Scholarship recipient

LEADERSHIP & ACHIEVEMENTS

Principal Investigator of computing time allocation grants, one at CSCS ALPS and one at GCS | 2025
Workshop Organizer – HPC & Quantum Computing workshop — ECT* | 2025
Invited Speaker – Scientific seminars at CERN and ECT* — 2025 – 2026
Visiting Research Grant – Open round 2026/1 fund — TU Wien | 2026